

A New Type of Quantum Exponential Speed Up: A Polynomial Time Quantum Algorithm For a Certain System of Modular Equations

M. Chowdhury

Written 26th August 2025 *

(Date of server time stamp)

Draft version.

Abstract

A modular equation is an equation that satisfies a congruence condition. We present a new quantum algorithm that can verify the existence of an exponential system of modular equations in polynomial time and polynomial space. The system of modular equations is of the form of a linear combinations of irrational numbers modulo 1.

1 Introduction

Note that all matrices are in standard notation unless otherwise stated. A modular equation is an equation that satisfies a congruence condition. We present a new quantum algorithm that can verify or solve the of an exponential system of modular equations in polynomial time and polynomial space(in its best running time). The system of modular equations is of the form of a linear combinations of irrational numbers modulo 1. For comparison, a previous major quantum algorithm for solving equations on a quantum computer is the HHL algorithm for solving linear equation with rational constants. Our new algorithm for solving works in a different way for a different type of system of equation-using irrationals instead of rationals and in a modular system of equations, our algorithm solves an exponential numbers of equations -not a linear number-in the input length of the number of bits. One possible disadvantage of the HHL algorithm is that it's running time is determined by how efficient the matrix b can be prepared, where the HHL algorithm computes the solution as $A^{-1}b$ in some form, this disadvantage is also mirrored in our algorithm -in other words the running time of our algorithm is determined by our corresponding matrix of constants b used. Another possible disadvantage of the HHL algorithm is that it does not

actually gives the specific values of the solution (i.e. the entries of the matrix $A^{-1}b$), in comparison, our algorithm does give the individual entries of the solution vector however our solution is exponentially slower than HHL algorithm-still our algorithm has an exponential speedup using a quantum computer of the corresponding classical version of the algorithm that solves the system of equation we solve.

Our algorithm is exactly described as follows.

Problem 1: solve a system of equations. Our algorithm does not involve a quantum way to do linear algebra to invert matrix as is done in the HHL algorithm. We write $F(x)$ for the fractional part x reduced modulo 2π i.e. $F(x) = x \bmod 2\pi$. Let $\gamma_1, \gamma_2, \dots, \gamma_n$ be irrational numbers of the form

$$\gamma_j = \gamma'_j n_j \bmod 2\pi = F(\gamma'_j)$$

where n_j is some positive integer and γ'_j is some irrational real number where $0 < j \leq n$

Define the set $S = \{\gamma_1, \gamma_2, \dots, \gamma_n\}$

For all distinct subset of $P_L \in S$ (of which there are 2^n such sets) denoted by $P_L = \{\gamma''_{j_1}, \gamma''_{j_2}, \dots, \gamma''_{j_k}\}$ for some integers $j_1, j_2, \dots, j_k$ we can test and thus solve (assuming the solution exists within the parameters of our algorithm for finding the solution)

$$\begin{aligned} & -\epsilon_L < (F(\gamma''_{j_1} + \gamma''_{j_2}, \dots, +\gamma''_{j_k}) \bmod 2\pi) - \alpha_L < \epsilon_L, \gamma''_a \text{ is equal to some } \gamma_b \text{ for some } a, b \\ & \text{for some real numbers } \alpha_L \text{ (known before hand) and some real number } \epsilon_L \\ & 1 \leq L \leq 2^n \end{aligned} \tag{5}$$

in other words for any element , we can solve the 2^n equations (where each equation corresponds to an element in the power set)

$$\begin{aligned}
F(\gamma''_{j_1}) - \alpha_{c_{j_1}} &= 0 \\
F(\gamma''_{j_2}) - \alpha_{c_{j_2}} &= 0 \\
F(\gamma''_{j_k}) - \alpha_{c_{j_k}} &= 0 \\
F(\gamma''_{j_1} + \gamma''_{j_2}) - \alpha_{c_{j_1 \cdot j_2}} &= 0 \\
F(\gamma''_{j_1} + \gamma''_{j_3}) - \alpha_{c_{j_1 \cdot j_3}} &= 0 \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_k}) - \alpha_{c_{j_1 \cdot j_k}} &= 0 \\
&\dots \\
F(\gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-1} \cdot j_k}} &= 0 \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_2} + \gamma''_{j_3}) - \alpha_{c_{j_1 \cdot j_2 \cdot j_3}} &= 0 \\
&\dots \\
F(\gamma''_{j_{k-2}} + \gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-2} \cdot \dots \cdot j_{k-1} \cdot j_k}} &= 0 \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k}) - \alpha_{c_{j_1 \cdot j_2 \cdot \dots \cdot j_k}} &= 0
\end{aligned} \tag{1}$$

by solving the system of inequalities

$$\begin{aligned}
|F(\gamma''_{j_1}) - \alpha_{c_{j_1}}| &< \epsilon_{j_1} \\
|F(\gamma''_{j_2}) - \alpha_{c_{j_2}}| &< \epsilon_{j_2} \\
|F(\gamma''_{j_k}) - \alpha_{c_{j_k}}| &< \epsilon_{j_k} \\
|F(\gamma''_{j_1} + \gamma''_{j_2}) - \alpha_{c_{j_1 \cdot j_2}}| &< \epsilon_{c_{j_1 \cdot j_2}} \\
|F(\gamma''_{j_1} + \gamma''_{j_3}) - \alpha_{c_{j_1 \cdot j_3}}| &< \epsilon_{c_{j_1 \cdot j_3}} \\
&\dots \\
|F(\gamma''_{j_1} + \gamma''_{j_k}) - \alpha_{c_{j_1 \cdot j_k}}| &< \epsilon_{c_{j_1 \cdot j_k}} \\
&\dots \\
|F(\gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-1} \cdot j_k}}| &< \epsilon_{c_{j_{k-1} \cdot j_k}} \\
&\dots \\
|F(\gamma''_{j_1} + \gamma''_{j_2} + \gamma''_{j_3}) - \alpha_{c_{j_1 \cdot j_2 \cdot j_3}}| &< \epsilon_{c_{j_1 \cdot j_2 \cdot j_3}} \\
&\dots \\
|F(\gamma''_{j_{k-2}} + \gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-2} \cdot j_{k-1} \cdot j_k}}| &< \epsilon_{c_{j_{k-2} \cdot j_{k-1} \cdot j_k}} \\
&\dots \\
|F(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k}) - \alpha_{c_{j_1 \cdot j_2 \cdot \dots \cdot j_k}}| &< \epsilon_{c_{j_1 \cdot j_2 \cdot \dots \cdot j_k}}
\end{aligned} \tag{1b}$$

where the "alpha constants" of the type $\alpha_{c_{j_1 \cdot j_2 \cdot \dots \cdot j_k}}$ are equal to some α_L depending on $j_1, j_2, \dots, j_k$ with some bound of the type $\epsilon_{c_{j_1 \cdot j_2 \cdot \dots \cdot j_k}}$ equal

to some ϵ_L -of course $j_1, j_2, \dots, j_k$ are different list of numbers, or value of L , for each element of the power set of S . Note that L is a label for the distinct subset of S therefore there is 2^n values of L . There is no polynomial time classical algorithm for solving the system of equations 1-one reason for this is because the constants on the right hand side are, in general, all unrelated and unique so there is no shortcut but to use each constant once thus this takes an exponential number of steps to use each constant to evaluate in an equation since all we know about the "alpha constants" is they are simply are 2^n real numbers/constants. In fact, it is easy to see from the system of equations 1, the "alpha constants" must have the certain form shown by each equation which is the left hand side of the system of equation 1. By the Weyl criterion there can never be an exact solution to the system of equation 1 but the Weyl Criterion ensures that for a large enough input search space for the solution, using a brute force search for the solution, *that a solution to the system of equation 1 always exists* with a desired accuracy, except for certain pathological cases with "artificially constructed irrationals" hence the only known algorithm, to solve the system of equation 1, is to enumerate all 2^n subsets and test them for equality-in other words a brute force classical algorithm. Note that a quantum gate belong to continuous symmetry groups, so in real hardware the epsilons will always be non zero. Let us now describe this brute force algorithm in more detail now we have described the systems of equation we are exactly going to solve. We will then describe a quantum version of this algorithm which quite significantly speeds up the classical brute force algorithm by an exponential factor regarding it's time complexity.

2 The Classical Brute Force Algorithms

This algorithm just tries every possible solution and stops if a solution is found otherwise it has failed.

Brute force algorithm (sketch).

Assume that the $1 \leq n_j \leq U$. U is just the bound for all possible solutions. As mentioned for a large enough integer U the solution exists by the Weyl criterion. The input to the algorithm is the "alpha constants" and the set of irrational constant γ'_h for some integers h .

Step 1) Select j integer values, in an ordered manner, for the 2^n integer values $n_1, n_2, \dots, n_{2^n}$

Step 2) Substitute the values into the system of equation 1, if the solution solves the system to the desired accuracy, depending on the epsilons given by the RHS of 1b, then jump to step 3 solution is found. If no solution was found then go to step one for a new candidate solu-

tion;but if all candidate solutions solutions have been tested, and they are all non-solution, then jump to step 4.

Step 3) Output solution, since a solution was found. Stop.

Step 4) Failed. No solutions found. Stop.

End of algorithm.

We now give a quantum version of this algorithm with a significant speed up. Since the speed up is from step 2 by an exponential amount using a quantum computer.

3 A New Quantum Algorithm 1: for Solving a System of Modular Equations Using a New Version of the Quantum Fourier Transform

We now describe the algorithm which uses our new version of the quantum Fourier transform. In a nutshell the algorithm works by using quantum properties to test an exponential number of equations in polynomial time, if the decision problem is true or false has an effect on the way phases interfere in the QFT, i.e. some decision problem is embedded into computing a local part of the QFT, this results in QFT finding a known period or not ,respectively.

Let F_{2^m} denote the QFT on the the integers domain 0 to $-1 + 2^m$.

The QFT is the following transform on $|X\rangle$

$$|X\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{Xk} |k\rangle$$

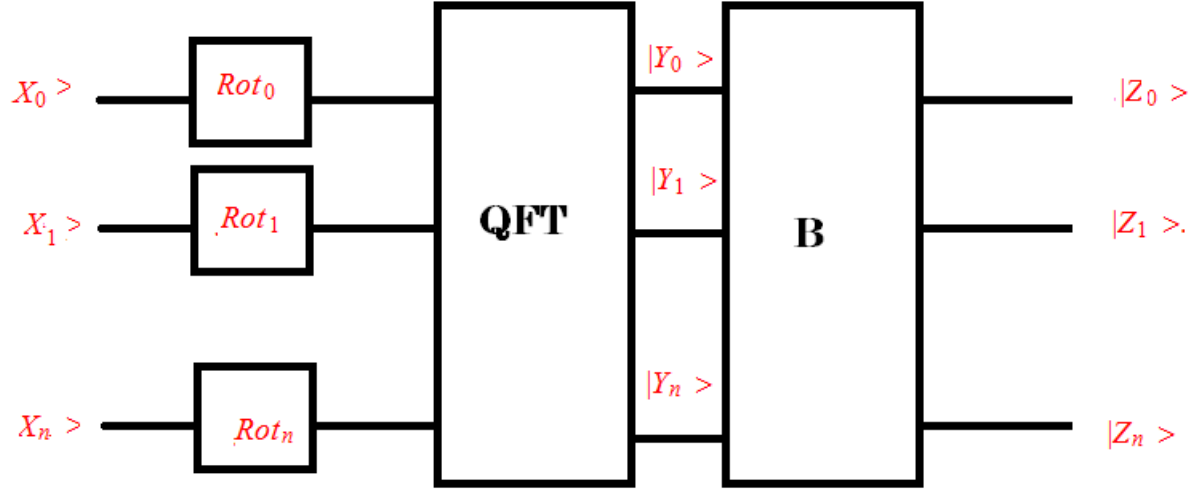
where $\omega = \frac{i2\pi}{N}$

which is a quantum version of the well known discrete Fourier transform which is derived from the integral version of the classical integral version of the Fourier transform-importantly this means that the QFT can be used to detect periodicity. Now,as is already known,the above can be factorised, when the domain is a power of 2, untangled qubits

$$|X\rangle = \left(\frac{|0\rangle + \omega^{-2^n X} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + \omega^{-2^{n-1} X} |1\rangle}{\sqrt{2}} \right) \otimes \dots \otimes \left(\frac{|0\rangle + \omega^{-X} |1\rangle}{\sqrt{2}} \right)$$

now we consider the following new variant of the QFT

$$|X\rangle = \left(\frac{|0\rangle + Rot_n \omega^{-2^n X} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + Rot_{n-1} \omega^{-2^{n-1} X} |1\rangle}{\sqrt{2}} \right) \otimes \dots \otimes \left(\frac{|0\rangle + Rot_0 \omega^{-X} |1\rangle}{\sqrt{2}} \right) \quad (2)$$



where

$$Rot_j = e^{i\gamma'_j n_j}$$

is a rotation gate which implement the left hand side of the equations 1.

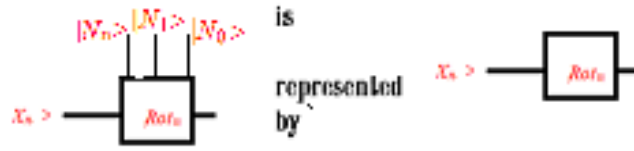
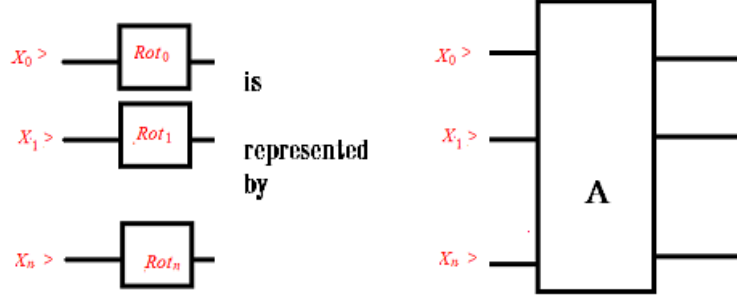
This can be represented by using the known polynomial circuit for the QFT, which is shown as the "middle circuit" in the diagram below, and for each input wire apply the unconditional rotation to represent each Rot_j for some integer j which is shown by the diagram below. The sequence of rotation gates applied in parallel with polynomial resources is referred to as A . Where the quantum binary expansion is $|X\rangle = |X_n X_{n-1} \dots X_0\rangle$ and the output which is our new version of the QFT is represented by $|Y_n Y_{n-1} \dots Y_0\rangle$ which is input to the unitary matrix described below, which has the final output $|Z_n Z_{n-1} \dots Z_0\rangle$.

The final output of the circuit is given by the qubits $|Z_n\rangle |Z_1\rangle \dots |Z_0\rangle$

where so the rotation gates/subcircuit have the input the integer

$N = |N_n\rangle |N_1\rangle \dots |N_0\rangle$ (not shown on RHS for short hand).

B is a matrix which encodes the values 2^n constants $\alpha_{c_{j_1} \dots j_k}$, shown on the RHS of 1, for clarity of explanation we take B to be as follows, which is a randomly matrix that can be constructed in exponential time and exponential space and probably has one of the simplest useful forms, but in general terms B may have quite a different form and at the end of this explantion we give another possible construction of B that takes



polynomial time and polynomial space. So B is a unitary matrix which has zero entries everywhere except the diagonal entries.

$$B = \begin{pmatrix} e^{-i\alpha_{c_{j_1}}} & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & e^{-i\alpha_{c_{j_2}}} & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & .00 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 0.0 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 00. & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & e^{-i\alpha_{c_{j_1} \cdot j_2}} & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & e^{-i\alpha_{c_{j_1} \cdot j_3}} & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & .000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0.00 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 00.0 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 000. & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & e^{-i\alpha_{c_{j_{k-2} \cdot \dots \cdot j_{k-1}, j_k}}} & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, .0 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 0. & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & e^{-i\alpha_{c_{j_1 \cdot j_2 \cdot \dots \cdot j_k}}} \end{pmatrix}$$

therefore the action of right multiplying our new version of the QFT

by the specific choice of B above gives

$$\begin{aligned}
 |X\rangle &\rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (\Pi_{i=0}^{u_i} Rot_{d_i} e^{-i\alpha_{h_i}}) \omega^{xk} |k\rangle \\
 &= \frac{1}{\sqrt{N}} (|0\rangle + e^{i(\gamma''_{j_1} - \alpha_{c_{j_1}})} \omega^x |1\rangle + \\
 &\quad e^{i(\gamma''_{j_2} - \alpha_{c_{j_2}})} \omega^{2x} |2\rangle + \\
 &\quad \dots + |e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k} - \alpha_{c_{j_1..j_2...j_k}})} \omega^{2X(N-1)} | (N-1) \rangle \\
 &\quad \text{where } \omega = \frac{i2\pi}{N} \text{ and } d_i, h_i \text{ is a sequence} \\
 &\quad \text{of } u_i \text{ integers}
 \end{aligned} \tag{9}$$

applies to periodic function in matrix for this new version of the QFT is as follows (informally it may be described as a "slightly wobbly QFT" or a QFT that has "wobbled" so much it behaves randomly depending on how good or bad the candidate solution is for the system of equations 1-in other words if the QFT is "stable" enough then the solution to the system of equation 1 is found)

$$W = \frac{1}{\sqrt{N}} \begin{pmatrix} 1 & e^{i(\gamma''_{j_1} - \alpha_{c_{j_1}})} & e^{i(\gamma''_{j_2} - \alpha_{c_{j_2}})} & \dots & e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k} - \alpha_{c_{j_1..j_2...j_k}})} \\ 1 & \omega e^{i(\gamma''_{j_1} - \alpha_{c_{j_1}})} & \omega^1 e^{i(\gamma''_{j_2} - \alpha_{c_{j_2}})} & \dots & \omega^{N-1} e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k} - \alpha_{c_{j_1..j_2...j_k}})} \\ 1 & \omega^2 e^{i(\gamma''_{j_1} - \alpha_{c_{j_1}})} & \omega^4 e^{i(\gamma''_{j_2} - \alpha_{c_{j_2}})} & \dots & \omega^{N-2} e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k} - \alpha_{c_{j_1..j_2...j_k}})} \\ \dots & \dots & \dots & \dots & \dots \\ \cdot & \dots & \cdot & \dots & \cdot \\ \cdot & \dots & \cdot & \dots & \cdot \\ \cdot & \dots & \cdot & \dots & \cdot \\ 1 & \omega^{N-1} e^{i(\gamma''_{j_1} - \alpha_{c_{j_1}})} & \omega^{N-2} e^{i(\gamma''_{j_2} - \alpha_{c_{j_2}})} & \dots & \omega^1 e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k} - \alpha_{c_{j_1..j_2...j_k}})} \end{pmatrix} \tag{9b}$$

thus the system of equations to solves is encoded in the amplitudes which form the vector in the dot product for each different ket that is transformed by the QFT. W represents our new version of the QFT note

that W can be decomposed into

$$W = W'QFT_n \text{ where } QFT_n \text{ is the (standard) QFT on the integers } |0\rangle, \dots, |-1 + 2^n\rangle$$

$$W' = \begin{pmatrix} e^{i(\gamma''_{j_1} - \alpha_{c_{j_1}})} & 0 & 0000000000, \dots, 00 & 0 \\ 0 & e^{i(\gamma''_{j_2} - \alpha_{c_{j_2}})} & 0000000000, \dots, 00 & 0 \\ 0 & 0 & .0000000000, \dots, 00 & 0 \\ 0 & 0 & 0.0000000000, \dots, 00 & 0 \\ 0 & 0 & 00.00000000, \dots, 00 & 0 \\ 0 & 0 & 000.000000, \dots, 00 & 0 \\ 0 & 0 & 0000.00000, \dots, 00 & 0 \\ 0 & 0 & 00000.0000, \dots, 00 & 0 \\ 0 & 0 & 000000.000, \dots, 00 & 0 \\ 0 & 0 & 0000000.00, \dots, 00 & 0 \\ 0 & 0 & 00000000.0, \dots, 00 & 0 \\ 0 & 0 & 000000000., \dots, 00 & 0 \\ 0 & 0 & 0000000000, \dots, 00 & 0 \\ 0 & 0 & 0000000000, \dots, .0 & 0 \\ 0 & 0 & 0000000000, \dots, 0. & 0 \\ 0 & 0 & 0000000000, \dots, 00 e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k} - \alpha_{c_{j_1, j_2, \dots, j_k}})} & 0 \end{pmatrix}$$

The decomposition of W as $W = W'QFT_n$ is can be used for building circuit that uses polynomial resources, using the following factorisation

$$A = \left(\frac{|0\rangle + Rot'_n |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + Rot'_{n-1} |1\rangle}{\sqrt{2}} \right) \otimes \dots \otimes \left(\frac{|0\rangle + Rot'_0 |1\rangle}{\sqrt{2}} \right) = W' B^{-1} \quad (10)$$

each term in 10 above has input of the integers n_j . Also the decomposition $W = W'QFT_n$ is useful for analysing the behaviour of the algorithm below (in step 7). Now measuring the state of above after using 9b, after applying it to a function of the form $x^i \bmod N$ (a periodic function which is used in the Shor algorithm) with a known period, after using the candidate solution, then recovering the value of the known period similar to how it is done in the Shor algorithm will recover the known period and thus this means the solution to the system of equations which is true otherwise it is false. We select various known period randomly and apply our transform 9b to these various known periods to recover them. So the new quantum algorithm to solve the system of equations is as follows

Repeat the following algorithm a number of times for different N where the factorisation of N is known

1) Pick a number $N = op$ where the factorisation of N is known for two known primes o and p pick a trandom nuner x such that $\gcd(x, N) =$

1 this means we can classically efficiently compute the period P function $f(i) = x^i \bmod N$ for fixed x .

2) Select the input domain size as some power of two q where $N^2 < q < 2N^2$

3) For the registers $|a\rangle |b\rangle$ to initially be $|a\rangle |b\rangle = |0\rangle |0\rangle$

4) Apply a tensor product of Hadamards to the first register $|a\rangle$ to obtain

$$\sum_{j=0}^q |j\rangle |0\rangle$$

5) Now apply the $U_x : |j\rangle |0\rangle \rightarrow |j\rangle |x^j \bmod N\rangle$ thus

$$\frac{1}{\sqrt{q}} \sum_{j=0}^q |j\rangle |x^j \bmod N\rangle$$

6) Now measure the second register this will result in a state of the form (if the period is a power of 2)

$$\frac{1}{\sqrt{q/p}} \sum_{j=0}^{-1+q/P} |ar + b_0\rangle |x^{b_0} \bmod N\rangle$$

(when the period is not a power of 2 the form of above changes slightly using to represent that the domain doesn't contain a whole number of periods)

7). Now apply B^{-1} then W^{-1} which is inverse of the new version of the quantum Fourier transform where B and W' represent the unitary operators described above

$$\begin{aligned} & \frac{1}{\sqrt{q/P}} W^{-1} B^{-1} \left(\sum_{j=0}^{-1+q/P} \left(\frac{1}{\sqrt{q}} \sum_{k=0}^{-1+q} \exp((-2\pi i k(ar + b_0))/q) |k\rangle \right) |x^{b_0} \bmod N\rangle = \right. \\ & \frac{1}{\sqrt{q/P}} W'^{-1} QFT_n^{-1} B^{-1} \left(\sum_{j=0}^{-1+q/P} \left(\frac{1}{\sqrt{q}} \sum_{k=0}^{-1+q} \exp((-2\pi i k(ar + b_0))/q) |k\rangle \right) |x^{b_0} \bmod N\rangle \right. \\ & \frac{1}{\sqrt{P}} (\exp(F(\gamma''_{j_1}) - \alpha_{c_{j_1}}) (\exp(2\pi * 0 * b_0/P) | \frac{0 * q}{P} \rangle + \\ & \exp(F(\gamma''_{j_2}) - \alpha_{c_{j_2}}) (\exp(2\pi * 1 * b_0/P) | \frac{1 * q}{P} \rangle + \dots \\ & + (F(\gamma''_{j_k}) - \alpha_{c_{j_k}}) (\exp(2\pi * k * b_0/P) | \frac{k * q}{P} \rangle + \dots \\ & \exp(F(\gamma''_{j_{k-2}} + \gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-2}, \dots, j_{k-1}, j_k}}) (\exp(2\pi * (-1 + P) * b_0/P) | \frac{(-1 + P) * q}{P} \rangle \\ & \left. \left. \right) | x^{b_0} \bmod N \rangle \right. \end{aligned}$$

if we have a solution for the system of equation 1 then the peaks where k is an integer multiple of the known inverse of the period P because the above simplifies to

$$\begin{aligned}
&= \frac{1}{\sqrt{P}} W^{-1} B^{-1} \left(\sum_{R=0}^{P-1} \exp(2\pi R b_0 / P) \left| \frac{Rq}{P} \right\rangle \right) |x^{b_0} \bmod N \rangle = \\
&\approx \frac{1}{\sqrt{P}} \left(\sum_{R=0}^{P-1} \exp(2\pi R b_0 / P) \left| \frac{Rq}{P} \right\rangle \right) |x^{b_0} \bmod N \rangle \text{ or} \\
&= \frac{1}{\sqrt{P}} \left(\sum_{R=0}^{P-1} \exp(2\pi R b_0 / P) \left| \frac{Rq}{P} \right\rangle \right) |x^{b_0} \bmod N \rangle
\end{aligned}$$

for a sufficiently good solution to the system of equations 1 in other words the above reduces to the known QFT transforming a periodic function.

$$\begin{aligned}
&F(\gamma''_{j_1}) - \alpha_{c_{j_1}} = 0 \\
&F(\gamma''_{j_2}) - \alpha_{c_{j_2}} = 0 \\
&F(\gamma''_{j_k}) - \alpha_{c_{j_k}} = 0 \\
&F(\gamma''_{j_1} + \gamma''_{j_2}) - \alpha_{c_{j_1, j_2}} = 0 \\
&F(\gamma''_{j_1} + \gamma''_{j_3}) - \alpha_{c_{j_1, j_3}} = 0 \\
&\dots \\
&F(\gamma''_{j_1} + \gamma''_{j_k}) - \alpha_{c_{j_1, j_k}} = 0 \\
&\dots \\
&F(\gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-1}, j_k}} = 0 \\
&\dots \\
&F(\gamma''_{j_1} + \gamma''_{j_2} + \gamma''_{j_3}) - \alpha_{c_{j_1, j_2, j_3}} = 0 \\
&\dots \\
&F(\gamma''_{j_{k-2}} + \gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-2}, \dots, j_{k-1}, j_k}} = 0 \\
&\dots \\
&F(\gamma''_{j_1} + \gamma''_{j_2} \dots + \gamma''_{j_k}) - \alpha_{c_{j_1, j_2, \dots, j_k}} = 0
\end{aligned} \tag{1}$$

We now sketch the proof of why this circuit works for verifying an exponential number of equations in polynomial time. Note that in general a reduction from a quantum search algorithm to a decision problem it may not be always true that the decision version can be done in polynomial time-but in this case it can as it used the structure of the equation 1 -rather than simply doing a search for the integers solutions.

3.1 Proof (sketch)

Suppose to simplify the sketch of the proof that B is an identity matrix of some size. Suppose that we have a solution for the systems of equation 1 thus we have that the condition 5 is satisfied thus 9 reduces as follows

$$|X\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (\Pi_{i=0}^{u_i} Rot_{d_i} e^{-i\alpha_{h_i}}) \omega^{xk} |k\rangle \quad (3)$$

$$\begin{aligned} &= \frac{1}{\sqrt{N}} (|0\rangle + e^{i\epsilon_0} \omega^x |1\rangle + e^{i\epsilon_1} \omega^{2x} |2\rangle + \dots + |e^{i\epsilon_N} \omega^{2X(N-1)}| (N-1)\rangle \\ &\approx \frac{1}{\sqrt{N}} (|0\rangle + \omega^x |1\rangle + \omega^{2x} |2\rangle + \dots + \omega^{2X(N-1)} | (N-1)\rangle \end{aligned} \quad (3b)$$

where $\omega = \frac{i2\pi}{N}$

in words -if all the complex amplitude episilons are close enough in magnitude to zero, then $e^{i\epsilon_0}, e^{i\epsilon_1}, \dots, e^{i\epsilon_N}$ is a good approximation where the vector of the dot product mentioned above is a sequence of 1's in the vector, thus equation 3 converges to 3b which is the QFT, thus the period can be successfully recovered, by the QFT, by sampling for the period a polynomial number of times. We can certainly say that if the episilons in $e^{i\epsilon_0}, e^{i\epsilon_1}, \dots, e^{i\epsilon_N}$ are small enough such that their collective error is smaller than the error tolerance of the real quantum gates, that represent the corresponding ideal quantum gates, then the period can be successfully recovered, by the QFT, by sampling for the period a polynomial number of times in these cases. It is known that generally the QFT still works if some of the small terms in it are ignored, here this may mean if one of the equations is nearly solved this has an effect on 2^n of the 2^{2n} coefficients in the QFT unitary matrix-but overall the QFT still works.

Now consider the case when we do not have a solution of the system of equations 1 then we have 9 reduces as the follows

$$|X\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (\Pi_{i=0}^{u_i} Rot_{d_i} e^{-i\alpha_{h_i}}) \omega^{xk} |k\rangle \quad (6)$$

$$\begin{aligned} &= \frac{1}{\sqrt{N}} (|0\rangle + e^{i\gamma''_{j_1}} \omega^x |1\rangle + e^{i\gamma''_{j_2}} \omega^{2x} |2\rangle + \dots + |e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k})} \omega^{2X(N-1)}| (N-1)\rangle \\ &\neq \frac{1}{\sqrt{N}} (|0\rangle + \omega^x |1\rangle + \omega^{2x} |2\rangle + \dots + \omega^{2X(N-1)} | (N-1)\rangle \end{aligned} \quad (6b)$$

where $\omega = \frac{i2\pi}{N}$

in words , the integers are not solutions then 6 does not converge to the QFT 6b. This is because by the Weyl Criterion the complex

phases $e^{i\gamma''_{j_1}}, e^{i\gamma''_{j_2}}, \dots, e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k})}$ make a very good approximation to a random of complex amplitudes-therefore 6b does not behave as a QFT but reveals no information about the period -the measurements will only give a random number. In other words a non-solution will not reveal the known period after a polynomial number of samplings. In other words when all the errors are small enough, most, an exponential number, of the QFT coefficient are, in a way, preserved, when an exponential number of the coefficients are preserved the QFT works, the QFT works because it uses an polynomial time algorithm while using the exponential numbers of QFT matrix entries are preserved, i.e. we can test the 2^n equations have a solutions by using a polynomial time and space algorithm which is the QFT for randomly selects periods.

We now give, in the next section, a related algorithm for solving a non-linear or linear systems of equation, at it heart, the algorithm uses the same idea, as the above algorithm, which is to embed the solution of a system of equations into the QFT resulting in a "stable" QFT or not, but it uses above "stability of QFT" idea in a different way which result is based on a different way to converge to the QFT.

4 A New Quantum Algorithm 2: for Solving a Related System of Modular Equations Using a New Version of the Quantum Fourier Transform

The algorithm below is a different algorithm that solves a different system of equations but in the "same family" of equations.

Repeat the following algorithm a number of times for different N where the factorisation of N is known

1) Pick a number $N = op$ where the factorisation of N is known for two known primes o and p pick a random number x such that $\gcd(x, N) = 1$ this means we can classically efficiently compute the period P function $f(i) = x^i \bmod N$ for fixed x .

2) Select the input domain size as some power of two q where $N^2 < q < 2N^2$

3) For the registers $|a\rangle |b\rangle$ to initially be $|a\rangle |b\rangle = |0\rangle |0\rangle$

4) Apply a tensor product of Hadamards to the first register $|a\rangle$ to obtain

$$\sum_{j=0}^q |j\rangle |0\rangle$$

5) Now apply the $U_x : |j\rangle \rightarrow |j\rangle |x^j \bmod N\rangle$ thus

$$\frac{1}{\sqrt{q}} \sum_{j=0}^q |j\rangle |x^j \bmod N\rangle$$

6) Now measure the second register this will result in a state of the form (if the period is a power of 2)

$$\frac{1}{\sqrt{q/p}} \sum_{j=0}^{-1+q/P} |ar + b_0\rangle |x^{b_0} \bmod N\rangle$$

(when the period is not a power of 2 the form of above changes slightly etc.)

7). Now consider the operator

$$|X\rangle = \left(\frac{|0\rangle + (Rot''_n)^{-2^n X} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + (Rot''_{n-1})^{-2^{n-1} X} |1\rangle}{\sqrt{2}} \right) \otimes \dots \otimes \left(\frac{|0\rangle + Rot_0 \omega^{-X} |1\rangle}{\sqrt{2}} \right) \quad (2a)$$

where the above is represented by the matrix transformation Q' and where

$$Rot''_i = e^{(i\gamma'_j n_j)}$$

and consider the matrix C which is implemented at worst with an exponential number of gates (similar to B the first algorithm in this way)

$$C = \begin{pmatrix} e^{-i\alpha_{c_{j_1}}} & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & e^{-i\alpha_{c_{j_2}}} & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & .00 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 0.0 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 00. & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & e^{-i\alpha_{c_{j_1} \cdot j_2}} & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & e^{-i\alpha_{c_{j_1} \cdot j_3}} & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & .000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0.00 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 00.0 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 000. & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & e^{-i\alpha_{c_{j_{k-2} \cdot \dots \cdot j_{k-1} \cdot j_k}}} & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, .0 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 0. & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & e^{-i\alpha_{c_{j_1 \cdot j_2 \cdot \dots \cdot j_k}}} \end{pmatrix}$$

thus the transformation $Q'C$ is represented by the matrix

$$V = \frac{1}{\sqrt{N}} \begin{pmatrix} (e^{i(\gamma''_{j_1} - \alpha_{c_{j_1}})})^0 \\ (e^{i(\gamma''_{j_1} - \alpha_{c_{j_1}})})^0 \\ (e^{i(\gamma''_{j_1} - \alpha_{c_{j_1}})})^0 \\ \dots \\ (e^{i(\gamma''_{j_1} + \gamma''_{j_2} - \alpha_{c_{j_1..j_2}})})^0 \\ \dots \\ (e^{i(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k} - \alpha_{c_{j_1..j_2..j_k}})})^0 \\ \text{continued} \\ (e^{i(\gamma''_{j_2} - \alpha_{c_{j_2}})})^0 \\ (e^{i(\gamma''_{j_2} - \alpha_{c_{j_2}})})^1 \\ (e^{i(\gamma''_{j_2} - \alpha_{c_{j_2}})})^2 \\ \dots \\ \dots \\ \cdot \\ \cdot \\ (e^{i(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k} - \alpha_{c_{j_1..j_2..j_k}})})^{2(N-1)} \\ \text{continued} \\ (e^{i(\gamma''_{j_3} - \alpha_{c_{j_3}})})^0 \\ (e^{i(\gamma''_{j_3} - \alpha_{c_{j_3}})})^2 \\ (e^{i(\gamma''_{j_3} - \alpha_{c_{j_3}})})^4 \\ \dots \\ \dots \\ \dots \\ \dots \\ (e^{i(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k} - \alpha_{c_{j_1..j_2..j_k}})})^{3(N-1)} \\ \text{continued} \\ \dots (e^{i(\gamma''_{j_1} + \gamma''_{j_2} - \alpha_{c_{j_1..j_2}})})^0 \dots (e^{i(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k} - \alpha_{c_{j_1..j_2..j_k}})})^0 \\ \dots \cdot \dots (e^{i(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k} - \alpha_{c_{j_1..j_2..j_k}})})^{N-1} \\ \dots \cdot \dots (e^{i(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k} - \alpha_{c_{j_1..j_2..j_k}})})^{2(N-1)} \\ \dots \cdot \dots \dots \\ \dots \cdot \dots \cdot \\ \dots \cdot \dots \cdot \\ \dots \cdot \dots \cdot \\ \dots \cdot \dots (e^{i(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k} - \alpha_{c_{j_1..j_2..j_k}})})^{(N-1)(N-1)} \end{pmatrix}$$

(the matrix shown is meant to be read as 1 continious matrix from top to bottom of the three blocks seperated by the word "continued" which actually is not a part of the matrix-the blocks have one or more column of the matrix when read from left to right-the first block has

one column vector, the second block has one columns vector, the third block has one columns vector, the fourth block has at least two column vectors, all the column vectors are composed of a generic number of rows-the notation is a bit uncoventional so this is why it is explained above)

Now apply the unitary matrix which corresponds to 2a converges to the QFT when each Rot''_n is a good approximation to some known root of unity specifically when

$$\begin{aligned}
\gamma''_{j_1} - \alpha_{c_{j_1}} &\approx 2\pi/N \\
\gamma''_{j_2} - \alpha_{c_{j_2}} &\approx 2\pi/N \\
\gamma''_{j_3} - \alpha_{c_{j_3}} &\approx 2\pi/N \\
\gamma''_{j_4} - \alpha_{c_{j_4}} &\approx 2\pi/N \\
&\dots \\
\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k} - \alpha_{c_{j_1..j_2...j_k}} &\approx 2\pi/N \\
&\dots \\
\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k} - \alpha_{c_{j_1..j_2...j_k}} &\approx 2\pi/N
\end{aligned}$$

where is a good enough approximations then $Q'C$ a good approximation for ther QFT (i.e. converges to the QFT for practical purposes), in a sense the matrix C corrects the matrix Q' so it becomes a better approximation to the QFT, thus we can solve the system of equation below related to the system of equations 1b (which is considered mod 2π)

$$\begin{aligned}
F(\gamma''_{j_1}) - \alpha_{c_{j_1}} &= 2\pi/N & (1b) \\
F(\gamma''_{j_2}) - \alpha_{c_{j_2}} &= 2\pi/N \\
F(\gamma''_{j_k}) - \alpha_{c_{j_k}} &= 2\pi/N \\
F(\gamma''_{j_1} + \gamma''_{j_2}) - \alpha_{c_{j_1..j_2}} &= 2\pi/N \\
F(\gamma''_{j_1} + \gamma''_{j_3}) - \alpha_{c_{j_1..j_3}} &= 2\pi/N \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_k}) - \alpha_{c_{j_1..j_k}} &= 2\pi/N \\
&\dots \\
F(\gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-1}..j_k}} &= 2\pi/N \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_2} + \gamma''_{j_3}) - \alpha_{c_{j_1..j_2..j_3}} &= 2\pi/N \\
&\dots \\
F(\gamma''_{j_{k-2}} + \gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-2}...j_{k-1},j_k}} &= 2\pi/N \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_2} \dots + \gamma''_{j_k}) - \alpha_{c_{j_1..j_2...j_k}} &= 2\pi/N
\end{aligned}$$

in other words, in this algorithm, the size of the approximate QFT contributes to what equation is solved as a rational multiple of π and if we want to solve 1b RHS as zero and not a rational multiple of π then it make be sufficient to take n as large as possible in the QFT since the error decreases exponentially for solving the case for RHS as n increases.

Thus this reduces to V being equivalent to the quantum Fourier transform.i.e. after $Q'C$ is applied to the state. which gives the following

$$\begin{aligned} \frac{1}{\sqrt{q/P}} \left(\sum_{j=0}^{-1+q/P} \left(\frac{1}{\sqrt{q}} \sum_{k=0}^{-1+q} \exp((-2\pi i k(ar + b_0))/q) |k\rangle \right) |x^{b_0} \bmod N\rangle = \right. \\ \left. \frac{1}{\sqrt{P}} \left(\sum_{R=0}^{P-1} \exp(2\pi i Rb_0/P) \left| \frac{Rq}{P} \right\rangle \right) |x^{b_0} \bmod N\rangle = \right. \end{aligned}$$

if we have a solution for the system of equation 1b then the peaks where k is an integer multiple of the known inverse of the period P because the above simplifies to

$$\begin{aligned} &= \frac{1}{\sqrt{P}} A \left(\sum_{R=0}^{P-1} \exp(2\pi i Rb_0/P) \left| \frac{Rq}{P} \right\rangle \right) |B|x^{b_0} \bmod N\rangle = \\ |\Phi\rangle &= \frac{1}{\sqrt{P}} \left(\sum_{R=0}^{P-1} \exp(2\pi i Rb_0/P) \left| \frac{Rq}{P} \right\rangle \right) |x^{b_0} \bmod N\rangle \end{aligned}$$

when

$$\begin{aligned} e^{(im\gamma'_n n_n + i\beta_{c_{j_n}})} &= \omega = e^{-2\pi im/N} \\ e^{(mi\gamma'_{n-1} n_{n-1} + i\beta_{c_{j_{n-1}}})} &= \omega = e^{-2\pi im/N} \\ &\dots \\ e^{(im\gamma'_0 n_0 + i\beta_{c_{j_0}})} &= \omega = e^{-2\pi im/N} \end{aligned}$$

implies

$$\begin{aligned} (\gamma'_n m n_n + \beta_{c_{j_n}}) \bmod 2\pi &= \frac{2\pi}{N} \\ (\gamma'_{n-1} m n_{n-1} + \beta_{c_{j_{n-1}}}) \bmod 2\pi &= \frac{2\pi}{N} \\ &\dots \\ (\gamma'_0 m n_0 + \beta_{c_{j_0}}) \bmod 2\pi &= \frac{2\pi}{N} \end{aligned} \tag{50}$$

in other words

$$Q'C^{-1}$$

converges to the QFT or is a good approximation to the QFT when the system of equations 50 has a good enough solution. In other words the specific phase constants in the QFT, for a particular size QFT, determine the accuracy of the solution to 50 if the RHS is considered as an approximation to zero (can be non zero though). Now applying this state to Φ can recover the period using a Shor type routine. Applying it to a number of different states for Φ means that we can test an exponential number of equation for solution to modular equation using a polynomial number of calls and thus have solved the modular system of equations, or a sufficiently good solution to the system of equations 1. In other words the above reduces to the known QFT transforming a periodic function.

We illustrate this example on a 2 qubit Fourier transform then 10 becomes the following

$$\begin{aligned}
|X\rangle &= \quad (11) \\
&= \left(\frac{|0\rangle + Rot'_1(X)|1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + Rot'_0(X)|1\rangle}{\sqrt{2}} \right) = \\
&= \frac{1}{2}(|00\rangle + Rot'_1(X)|10\rangle + Rot'_0(X)|01\rangle + Rot'_1(X)Rot'_0(X)|11\rangle + \\
&+ \frac{1}{2}(|00\rangle + \exp(i * f_2(x_{0,s}, x_{2,s}, \dots, x_{z,s}, a_{0,s}, a_{2,s}, \dots, a_{z,s}))|10\rangle + \\
&+ \exp(i * f_1(x_{0,s}, x_{2,s}, \dots, x_{z,s}, a_{0,s}, a_{2,s}, \dots, a_{z,s}))|01\rangle + \\
&+ \exp(i * f_2(x_{0,s}, x_{2,s}, \dots, x_{z,s}, a_{0,s}, a_{2,s}, \dots, a_{z,s}))|11\rangle
\end{aligned}$$

choose $f_s(x_0, x_1, \dots, x_z, a_1, a_2, \dots, a_z) = (\sum_{T=1}^{T=z} ((sx_{0,s})^T a_{T,s} \bmod 2\pi)$
i.e. it has only one free variable and thus the state

(101)

$$\begin{aligned}
&= \frac{1}{2}(|00\rangle + \exp(i * f_2(x_{0,s}, a_{0,s}, a_{2,s}, \dots, a_{z,s}))|10\rangle + \\
&+ \exp(i * f_1(x_{0,s}, a_{0,s}, a_{2,s}, \dots, a_{z,s}))|01\rangle + \exp(i * f_2(x_{0,s}, a_{0,s}, a_{2,s}, \dots, a_{z,s}))|11\rangle
\end{aligned} \quad (102)$$

this transforms the state $|x_{0,s}\rangle$ by the unitary matrix

$$\frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ \text{continued} \\ 1 \\ \exp(f_1(x_{0,0}, a_{1,0}, a_{2,0}, \dots, a_{z,0})i) \\ \exp(f_1(2x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1})i) \\ \exp(f_1(3x_{0,2}, a_{1,2}a_{2,2}, \dots, a_{z,2})i) \\ \text{continued} \\ 1 \\ \exp(f_2(2x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1})i) \\ \exp(f_2(4x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1})i) \\ \exp(f_2(6x_{0,1}, a_{1,1}a_{2,1}, \dots, a_{z,1})i) \\ \text{continued} \\ 1 \\ \exp(f_1(x_{0,0}, a_{1,0}, a_{2,0}, \dots, a_{z,0}) + f_2(2x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1})i) \\ \exp(f_1(2x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1}) + f_2(4x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1})i) \\ \exp(f_1(9x_{0,2}, a_{1,2}a_{2,2}, \dots, a_{z,2}) + f_2(18x_{0,1}, a_{1,1}a_{2,1}, \dots, a_{z,1})i) \end{pmatrix}$$

Again, similar to the comment for the matrix above, the above is a 4 by 4 matrix meant to be read from left to right, and "continued" is actually not part of the matrix-but here the column vectors are give one by one -and are "stuck together" side by side from left to right to give the 4 by 4 matrix our notation expresses-this is explanation for non-standard notation. Note that the power of the exponential function in the matrix above are uniformly distributed modulo 1 which means the above matrix for some values will be a good approximation to, the 2 qubit QFT given below

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 \exp(i * \frac{1}{2}\pi) & \exp(i * 1\pi) & \exp(i * \frac{3}{2}\pi) & \\ 1 \exp(i * 1\pi) & \exp(i * 2\pi) & \exp(i * 3\pi) & \\ 1 \exp(i * \frac{3}{2}\pi) & \exp(i * 3\pi) & \exp(i * \frac{9\pi}{2}) & \end{pmatrix}$$

when

$$\begin{aligned}
f_1(x_{0,0}, a_{1,0}, a_{2,0}, \dots, a_{z,0}) \bmod 2\pi &= \frac{\pi}{2} (12) \\
f_2(2x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1}) \bmod 2\pi &= \pi \\
f_1(x_{0,0}, a_{1,0}, a_{2,0}, \dots, a_{z,0}) + f_2(2x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1}) \bmod 2\pi &= \frac{3}{2}\pi \\
f_1(2x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1}) \bmod 2\pi &= \pi \\
f_2(4x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1}) \bmod 2\pi &= 0 \\
f_1(2x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1}) + f_2(4x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1}) \bmod 2\pi &= \pi \\
f_1(3x_{0,2}, a_{1,2}, a_{2,2}, \dots, a_{z,2}) \bmod 2\pi &= \frac{3}{2}\pi \\
f_2(6x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1}) \bmod 2\pi &= \pi \\
f_1(9x_{0,2}, a_{1,2}, a_{2,2}, \dots, a_{z,2}) + f_2(18x_{0,1}, a_{1,1}, a_{2,1}, \dots, a_{z,1}) \bmod 2\pi &= \frac{\pi}{2}
\end{aligned}$$

where the equals, in the above system of equation 12, represent equality in a machine precision sense (or close enough in a practical implementation of a quantum computer). Then it is equivalent to the QFT to same precision, if the precision is close enough then it will recover the period of a periodic function using a Shor type procedure, thus it means that for all the free parameters it verifies the solution in polynomial resources -because all the entries of the QFT have to be approximated correctly to recover the randomly chosen periods-this means in polynomial time we can solve a set of $2^n * 2^n$ exponential equations parametrized by an integer between 1 and 2^n where the coefficients in the constant term are derives from the entries of some size of QFT (or another polynomial time transform).

This shows how solve a set of a number exponential number of $O(2^D)$ equations where the set is composed on linear combination of a set of size D .

4.1 A Related Problem

In this subsection we will now describe another algorithm that solves the following problem exactly described as follows-before giving a proof of the above.

Problem 2: solve a system of equations. Our algorithm does not involve a quantum way to do linear algebra to invert matrix as is done in the HHL algorithm. We write $F_M(x)$ for the fractional part and is any polynomial time computable function of n and m , x i.e. the simplest example is $F_M(x) = mx \bmod 2\pi$. Let $\gamma_1, \gamma_2, \dots, \gamma_n$ be irrational

numbers of the form

$$\gamma_j = m\gamma'_j n_j \bmod 2\pi = F_M(\gamma'_j)$$

a more complicated example in m is any function which behaves like a random number generator, in a statistical sense, in n or m e.g.

$$\gamma_j = RNG(\gamma'_j, n_j, m) \bmod 2\pi = F_M(\gamma'_j)$$

where

$$RNG(\gamma'_j, n_j, m) = \ln ||\gamma'_j n_j m| \bmod 1| \bmod 2\pi$$

where n_j is some positive integer and γ'_j is some irrational real number where $0 < j \leq n$

$$\text{Define the set } S = \{\gamma_1, \gamma_2, \dots, \gamma_n\}$$

m is an integer $1 \leq m \leq 2^n$ so problem 2 is similar to problem 1 except we are solving 2^{2^n} different equations which is a set of 2^n equation parameterised by m to given 2^n sets of 2^n equations i.e. 2^n equations. This version of $F_M(\gamma'_j)$ using $RNG(\gamma'_j, n_j, m)$ is implemented using the operator ω which has a similar role to W given above.

For all distinct subset of $P_L \in S$ (of which there are 2^n such sets) denoted by $P_L = \{\gamma''_{j_1}, \gamma''_{j_2}, \dots, \gamma''_{j_k}\}$ for some integers $j_1, j_2, \dots, j_k$ we can test and thus solve (assuming the solution exists within the parameters of our algorithm for finding the solution)

$$- \epsilon_L < (F_M(\gamma''_{j_1} + \gamma''_{j_2}, \dots, +\gamma''_{j_k}) \bmod 2\pi) - \alpha_L < \epsilon_L \quad (5)$$

for some real numbers α_L (known before hand) and some real number ϵ_L

$$1 \leq L \leq 2^n$$

in other words for any element , we can solve the 2^n equations (where each equation corresponds to an element in the power set)

$$\begin{aligned}
F_M(\gamma''_{j_1}) - \alpha_{c_{j_1}} &= 0 \\
F_M(\gamma''_{j_2}) - \alpha_{c_{j_2}} &= 0 \\
F_M(\gamma''_{j_k}) - \alpha_{c_{j_k}} &= 0 \\
F_M(\gamma''_{j_1} + \gamma''_{j_2}) - \alpha_{c_{j_1, j_2}} &= 0 \\
F_M(\gamma''_{j_1} + \gamma''_{j_3}) - \alpha_{c_{j_1, j_3}} &= 0 \\
&\dots \\
F_M(\gamma''_{j_1} + \gamma''_{j_k}) - \alpha_{c_{j_1, j_k}} &= 0 \\
&\dots \\
F_M(\gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-1}, j_k}} &= 0 \\
&\dots \\
F_M(\gamma''_{j_1} + \gamma''_{j_2} + \gamma''_{j_3}) - \alpha_{c_{j_1, j_2, j_3}} &= 0 \\
&\dots \\
F_M(\gamma''_{j_{k-2}} + \gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-2}, \dots, j_{k-1}, j_k}} &= 0 \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k}) - \alpha_{c_{j_1, j_2, \dots, j_k}} &= 0
\end{aligned} \tag{1}$$

by solving the system of inequalities

$$\begin{aligned}
|F_M(\gamma''_{j_1}) - \alpha_{c_{j_1}}| &< \epsilon_{j_1} \\
|F_M(\gamma''_{j_2}) - \alpha_{c_{j_2}}| &< \epsilon_{j_2} \\
|F_M(\gamma''_{j_k}) - \alpha_{c_{j_k}}| &< \epsilon_{j_k} \\
|F_M(\gamma''_{j_1} + \gamma''_{j_2}) - \alpha_{c_{j_1, j_2}}| &< \epsilon_{c_{j_1, j_2}} \\
|F_M(\gamma''_{j_1} + \gamma''_{j_3}) - \alpha_{c_{j_1, j_3}}| &< \epsilon_{c_{j_1, j_3}} \\
&\dots \\
|F_M(\gamma''_{j_1} + \gamma''_{j_k}) - \alpha_{c_{j_1, j_k}}| &< \epsilon_{c_{j_1, j_k}} \\
&\dots \\
|F_M(\gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-1}, j_k}}| &< \epsilon_{c_{j_{k-1}, j_k}} \\
&\dots \\
|F_M(\gamma''_{j_1} + \gamma''_{j_2} + \gamma''_{j_3}) - \alpha_{c_{j_1, j_2, j_3}}| &< \epsilon_{c_{j_1, j_2, j_3}} \\
&\dots \\
|F_M(\gamma''_{j_{k-2}} + \gamma''_{j_{k-1}} + \gamma''_{j_k}) - \alpha_{c_{j_{k-2}, j_{k-1}, j_k}}| &< \epsilon_{c_{j_{k-2}, j_{k-1}, j_k}} \\
&\dots \\
|F(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k}) - \alpha_{c_{j_1, j_2, \dots, j_k}}| &< \epsilon_{c_{j_1, j_2, \dots, j_k}}
\end{aligned} \tag{1b}$$

are some constant of the type $\alpha_{c_{j_1 \cdot j_2 \dots j_k}}$ (equal to some α_L) depending on $j_1, j_2, \dots, j_k$ with some bound of the type $\epsilon_{c_{j_1 \cdot j_2 \dots j_k}}$ equal to some ϵ_L - of course $j_1, j_2, \dots, j_k$ are different list of numbers, or value of L , for each element of the power set of S . Note that L is a label for the distinct subset of S therefore there is 2^n values of L . There is no polynomial time classical algorithm for solving the system of equations 1-one reason for this is because the constants on the right hand side are, in general, all unrelated and unique so there is no shortcut but to use each constant once thus this takes an exponential number of steps to use each constant to evaluate in an equation since all we know about the "alpha constants" is they are simply are 2^n real numbers/constants. By the Weyl criterion there can never be an exact solution to the system of equation 1 but the Weyl Criterion ensures that for a large enough input search space for the solution, using a brute force search for the solution, *that a solution to the system of equation 1 always exists* with a desired accuracy, except for certain pathological cases with "artificially constructed irrationals" hence the only known algorithm, to solve the system of equation 1, is to enumerate all 2^n subsets and test them for equality-in other words a brute force classical algorithm. Note that quantum gate belong to continous symmetry groups, so in real hardware the episolons will be non zero.

4.2 Proof (sketch)

Now back to the other section for the proof. Suppose that we have a solution for the systems of equation 1 thus we have that the condition 5, above given in the previous subsection, is satisfied thus 2a reduces as follows

$$|X\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (\Pi_{i=0}^{u_i} Rot_{d_i} e^{-i\alpha_{h_i}}) \omega^{xk} |k\rangle \quad (3)$$

$$\begin{aligned} &= \frac{1}{\sqrt{N}} (|0\rangle + e^{i\epsilon_0} \omega^x |1\rangle + e^{i\epsilon_1} \omega^{2x} |2\rangle + \dots + |e^{i\epsilon_N} \omega^{2X(N-1)}|(N-1)\rangle \\ &\approx \frac{1}{\sqrt{N}} (|0\rangle + \omega^x |1\rangle + \omega^{2x} |2\rangle + \dots + \omega^{2X(N-1)}|(N-1)\rangle \end{aligned} \quad (3b)$$

where $\omega = \frac{i2\pi}{N}$

in words -if all the complex amplitude episilons are close enough in magnitude to zero, then $e^{i\epsilon_0}, e^{i\epsilon_1}, \dots, e^{i\epsilon_N}$ is a good approximation where the vector of the dot product mentioned above is a sequence of 1's in the vector, thus equation 3 converges to 3b which is the QFT, thus the

period can be successfully recovered, by the QFT, by sampling for the period a polynomial number of times. We can certainly say that if the epsilons in $e^{i\epsilon_0}, e^{i\epsilon_1}, \dots, e^{i\epsilon_N}$ are small enough such that their collective error is smaller than the error tolerance of the real quantum gates, then the period can be successfully recovered, by the QFT, by sampling for the period a polynomial number of times in these cases.

Now consider the case when we do not have a solution of the system of equations 1 then we have 9 reduces as the follows

$$|X\rangle \rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} (\Pi_{i=0}^{u_i} Rot_{d_i} e^{-i\alpha_{h_i}}) \omega^{xk} |k\rangle \quad (6)$$

$$\begin{aligned} &= \frac{1}{\sqrt{N}} (|0\rangle + e^{i\gamma''_{j_1}} \omega^x |1\rangle + e^{i\gamma''_{j_2}} \omega^{2x} |2\rangle + \dots + |e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k})} \omega^{2X(N-1)} | (N-1) \rangle \\ &\neq \frac{1}{\sqrt{N}} (|0\rangle + \omega^x |1\rangle + \omega^{2x} |2\rangle + \dots + \omega^{2X(N-1)} | (N-1) \rangle \end{aligned} \quad (6b)$$

where $\omega = \frac{i2\pi}{N}$

in words, the integers are not solutions of 6 does not converge to the QFT 6b. This is because by the Weyl Criterion the complex phases $e^{i\gamma''_{j_1}}, e^{i\gamma''_{j_2}}, \dots, e^{i(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k})}$ make a very good approximation to a random of complex amplitudes-therefore 6b does not behave as a QFT but reveals no information about the period -the measurements will only give a random number. In other words a non-solution will not reveal the known period after a polynomial number of samplings. The solution is tested using the equations 1f.

$$\begin{aligned}
F(\gamma''_{j_1}) &= ?\alpha_{c_{j_1}} \\
F(\gamma''_{j_2}) &= ?\alpha_{c_{j_2}} \\
F(\gamma''_{j_k}) &= ?\alpha_{c_{j_k}} \\
F(\gamma''_{j_1} + \gamma''_{j_2}) &= ?\alpha_{c_{j_1 \cdot j_2}} \\
F(\gamma''_{j_1} + \gamma''_{j_3}) &= ?\alpha_{c_{j_1 \cdot j_3}} \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_k}) &= ?\alpha_{c_{j_1 \cdot j_k}} \\
&\dots \\
F(\gamma''_{j_{k-1}} + \gamma''_{j_k}) &= ?\alpha_{c_{j_{k-1} \cdot j_k}} \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_2} + \gamma''_{j_3}) &= ?\alpha_{c_{j_1 \cdot j_2 \cdot j_3}} \\
&\dots \\
F(\gamma''_{j_{k-2}} + \gamma''_{j_{k-1}} + \gamma''_{j_k}) &= ?\alpha_{c_{j_{k-2} \cdot \dots \cdot j_{k-1} \cdot j_k}} \\
&\dots \\
F(\gamma''_{j_1} + \gamma''_{j_2}, \dots, + \gamma''_{j_k}) &= ?\alpha_{c_{j_1 \cdot j_2 \cdot \dots \cdot j_k}}
\end{aligned} \tag{1f}$$

Algorithm(sketch): The steps of the algorithm are as follows.

Step 1) Decide on a value ρ an integer value which is the value of the periodic length we will attempt to construct later

Step 2) We have a quantum computer with the two registers $|x\rangle |y\rangle = |0\rangle |0\rangle$

Apply the transformation

$$|x\rangle |y\rangle \rightarrow H^{\otimes 2n} |0\rangle |0\rangle \rightarrow \frac{1}{\sqrt{2^{rega}}} \sum_{j=0}^{2^{rega}-1} |j\rangle |0\rangle = |\Psi\rangle$$

Step 3) Now apply P_x which is the unitary transform that performs the known periodic function with a known period which can be a primitive root

$$P_x |\Psi\rangle = \frac{1}{\sqrt{2^{rega}}} \sum_{j=0}^{2^{rega}-1} |j\rangle |x^j \bmod N\rangle$$

Step 4) Now apply the transform F_{2^m} then ω

Step 5) Measure the registers to compute the period using continued fractions

Step 6) if it is same as expected period then the criterion is satisfied otherwise not

End of (sketched) algorithm.

5 A New Quantum Algorithm 3: for Solving a Related System of Modular Equations Using a New Version of the Quantum Fourier Transform

The following is a third algorithm different from the other two above, it solves a different systems of equations, but is in the "same family" of equations that are solved

Algorithm 3 (sketch)

1) Pick a number $N = pq$ where the factorisation of N is known for two known primes p and q pick a random number x such that $\gcd(x, N) = 1$ this means we can classically efficiently compute the period P function $f(i) = x^i \bmod N$ for fixed x .

2) Select the input domain size as some power of 2 q where $N^2 < q < 2N^2$

3) For the registers $|a\rangle |b\rangle$ to initially be $|a\rangle |b\rangle = |0\rangle |0\rangle$

4) Apply a tensor product of Hadamards to the first register $|a\rangle$ to obtain

$$\sum_{j=0}^q |j\rangle |0\rangle$$

5) Now apply the $U_x : |j\rangle |0\rangle \rightarrow |j\rangle |x^j \bmod N\rangle$ thus

$$\frac{1}{\sqrt{q}} \sum_{j=0}^q |j\rangle |x^j \bmod N\rangle$$

6) Now measure the second register this will result in a state of the form (if the period is a power of 2)

$$\frac{1}{\sqrt{q/p}} \sum_{j=0}^{-1+q/P} |ar + b_0\rangle |x^{b_0} \bmod N\rangle$$

(when the period is not a power of 2 the form of above changes slightly using to represent that the domain does not contain a whole number of periods)

7). Now apply the unitary matrix corresponds to 10 , when it is a good approximation for the QFT, then this reduced to only applying quantum Fourier transform represented by

$$\begin{aligned} \frac{1}{\sqrt{q/P}} \left(\sum_{j=0}^{-1+q/P} \left(\frac{1}{\sqrt{q}} \sum_{k=0}^{-1+q} \exp((-2\pi i k(ar + b_0))/q) |k\rangle \right) |x^{b_0} \bmod N\rangle = \right. \\ \left. \frac{1}{\sqrt{P}} A \left(\sum_{R=0}^{P-1} \exp(2\pi i R b_0 / P) \left| \frac{Rq}{P} \right\rangle \right) |B|x^{b_0} \bmod N\rangle = \right. \end{aligned}$$

if we have a solution for the system of equation 1 then the peaks where k is an integer multiple of the known inverse of the period P because the above simplifies to

$$\begin{aligned} &= \frac{1}{\sqrt{P}} A \left(\sum_{R=0}^{P-1} \exp(2\pi i R b_0 / P) \left| \frac{Rq}{P} \right\rangle \right) |B|x^{b_0} \bmod N\rangle = \\ |\Phi\rangle &= \frac{1}{\sqrt{P}} \left(\sum_{R=0}^{P-1} \exp(2\pi i R b_0 / P) \left| \frac{Rq}{P} \right\rangle \right) |x^{b_0} \bmod N\rangle \end{aligned}$$

Now consider

$$|X\rangle = \left(\frac{|0\rangle + (Rot''_n)^{-2^n X} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + (Rot''_{n-1})^{-2^{n-1} X} |1\rangle}{\sqrt{2}} \right) \otimes \dots \otimes \left(\frac{|0\rangle + Rot_0 \omega^{-X} |1\rangle}{\sqrt{2}} \right) \quad (200)$$

this is a specialisation of

$$\begin{aligned} |X\rangle &= \left(\frac{|0\rangle + (Rot'''_n(x))^{g(n,X)} |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle + (Rot'''_{n-1}(x))^{g(n-1,X)} |1\rangle}{\sqrt{2}} \right) \otimes \\ &\dots \otimes \left(\frac{|0\rangle + Rot'''_0 \omega^{g(0,X)} |1\rangle}{\sqrt{2}} \right) \end{aligned} \quad (201)$$

where the above is represented by the matrix transformation Q' and where- for a concrete example for $g(n-1, X)$

$$Rot'''_i = (e^{(F_m(i\gamma'_j n_j))}) \text{ where for example} \quad (203)$$

$$(204)$$

$g(n-1, X) = \eta 2^{n+(X + floor((1/\cos(nX\zeta))^2) + c) \bmod 1}$ for some irrational η, ζ

$n + (X + floor((1/\cos(nX\zeta))^2) + c$ behaves like a random function with the parameters η, ζ, X, c by the Weyl criterion there exists constants

η, ζ, c such that $\eta 2^{n+(X+\text{floor}((1/\cos(nX\zeta))^2)+c)}$ will be $2^{-n}X$ will be a near approximation to $\eta 2^{n+(X+\text{floor}((1/\cos(nX\zeta))^2)+c)}$ for the above 3 constants for a range of values η, X this means we can solve the system of equations (using our algorithm) when we have the conditions

$$F'_M(i\gamma'_j n_j, X) = F_M(i\gamma'_j n_j) * g(n, X) = 2\pi R \text{ or } \approx 2\pi R \quad (248)$$

for some R rational which is an entry in QFT

where $(Rot'''_n(x))^{g(n,X)}$ is some constant entry in QFT matrix

define $X = n_j^2$ for a concrete example, then

$$\begin{aligned} F''_M(i\gamma'_j n_j) &= \\ F'_M(i\gamma'_j n_j, n_j^2) &= \\ F_M(i\gamma'_j n_j) * g(n, X) &= \\ F_m(i\gamma'_j n_j) * g(n, n_j^2) & \end{aligned} \quad (249)$$

then we can solve the following system of equation

$$\begin{aligned} F''_M(\gamma''_{j_1}) - \beta_{c_{j_1}} &= 0 \\ F''_M(\gamma''_{j_2}) - \beta_{c_{j_2}} &= 0 \\ F''_M(\gamma''_{j_k}) - \beta_{c_{j_k}} &= 0 \\ F''_M(\gamma''_{j_1}) + F''_M(\gamma''_{j_2}) - \beta_{c_{j_1 \cdot j_2}} &= 0 \\ F''_M(\gamma''_{j_1}) + F''_M(\gamma''_{j_3}) - \beta_{c_{j_1 \cdot j_3}} &= 0 \\ &\dots \\ F''_M(\gamma''_{j_1}) + F''_M(\gamma''_{j_k}) - \beta_{c_{j_1 \cdot j_k}} &= 0 \\ &\dots \\ F''_M(\gamma''_{j_{k-1}}) + F''_M(\gamma''_{j_k}) - \beta_{c_{j_{k-1} \cdot j_k}} &= 0 \\ &\dots \\ F''_M(\gamma''_{j_1}) + F''_M(\gamma''_{j_2}) + F''_M(\gamma''_{j_3}) - \beta_{c_{j_1 \cdot j_2 \cdot j_3}} &= 0 \\ &\dots \\ F''_M(\gamma''_{j_{k-2}}) + F''_M(\gamma''_{j_{k-1}}) + F''_M(\gamma''_{j_k}) - \beta_{c_{j_{k-2} \cdot \dots \cdot j_{k-1} \cdot j_k}} &= 0 \\ &\dots \\ F(\gamma''_{j_1} + \gamma''_{j_2} + \dots + \gamma''_{j_k}) - \beta_{c_{j_1 \cdot j_2 \cdot \dots \cdot j_k}} &= 0 \end{aligned} \quad (250)$$

with

$$C = \begin{pmatrix} e^{-i\beta_{c_{j_1}}} & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & e^{-i\beta_{c_{j_2}}} & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & .00 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 0.0 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 00. & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & e^{-i\beta_{c_{j_1} \cdot j_2}} & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & e^{-i\beta_{c_{j_1} \cdot j_3}} & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & .000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0.00 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 00.0 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 000. & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & e^{-i\beta_{c_{j_k-2} \dots j_{k-1}, j_k}} & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, .0 & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 0. & 0 \\ 0 & 0 & 000 & 0 & 0 & 0000 & 0 & , \dots, 00 & e^{-i\beta_{c_{j_1} \cdot j_2 \dots j_k}} \end{pmatrix}$$

we have a solution when

$$\begin{aligned} e^{iF_m(i\gamma'_n n_j) * g(\beta_{c_{j_n}}, X)} &= \omega = e^{-2\pi i m / N} \\ e^{iF_m(i\gamma'_{n-1} n_{n-1}) * g(\beta_{c_{j_{n-1}}}, X)} &= \omega = e^{-2\pi i m / N} \\ &\dots \\ e^{i(F_m(i\gamma'_0 n_0) * g(\beta_{c_{j_0}}, X))} &= \omega = e^{-2\pi i m / N} \end{aligned}$$

which implies

$$\begin{aligned} (F_m(i\gamma'_n n_j) * g(\beta_{c_{j_n}}, X)) \bmod 2\pi &= \frac{2\pi}{N} \\ (F_m(i\gamma'_{n-1} n_{n-1}) * g(\beta_{c_{j_{n-1}}}, X)) \bmod 2\pi &= \frac{2\pi}{N} \\ &\dots \\ (F_m(i\gamma'_0 n_0) * g(\beta_{c_{j_0}}, X)) \bmod 2\pi &= \frac{2\pi}{N} \end{aligned} \tag{55}$$

in other words

$$Q' C^{-1}$$

converges to the or is a good approximation to the QFT when the system of equations 55 has a good/close enough solution. In other words the specific phase constants in the QFT, for a particular size QFT, determine the accuracy of the solution to 50 if the RHS is considered as

an approximation to zero (can be non zero though). Now applying this operator ,to a fuction of known periods, can recover this period using a Shor related continued fraction sub/routine.Applying it to a number of different states/periods for means that we can test an exponential number of equation for solution to modular equation using a polynomial number of calls and thus have solved the modular system of equations for a sufficiently good solution to the system of equations above. In other words the above reduces to the known QFT tranforming a periodic function. This gives an generalised version of the above algorithm in the previous section. The proof is omitted but it has similarities to the proof in the above section.

6 References

Locate references